

A Report on  
**MACHINE LEARNING BASED PREDICTION AND REAL-TIME  
FORECASTING  
OF LIGHTNING STRIKES IN JHARKHAND**

*Submitted to DST PURSE Project 2022*

*Center for Advanced Research on Machine Learning for  
Geospatial Analytics and Solutions for Jharkhand State  
(ARMLEGS)*



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## **APPROVAL OF THE GUIDE**

Recommended that the Summer Internship Project titled submitted **MACHINE LEARNING BASED PREDICTION AND REAL-TIME FORECASTING OF LIGHTNING STRIKES IN JHARKHAND** by **Ankit Raj Tiwary, Roll No.: BTECH/60061/23, Birla institute of Technology Mesra off campus Deoghar** is approved by me for submission. To the best of my knowledge, the report represents work carried out by the student in **Birla Institute of Technology, Mesra** and the content of this report is not forming a basis for the award of any previous degree to anyone else.

**Date:23/07/2026**

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## **DECLARATION CERTIFICATE**

I certify that

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- The work has not been submitted to any other Institute for any other degree or diploma.
- I have followed the guidelines provided by the Institute in writing the report.
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- Whenever I have used materials (data, theoretical analysis, and text) from other sources, I have given due credit to them by citing them in the text of the report and giving their details in the references.
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## ***ABSTRACT***

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Jharkhand faces some of the highest lightning-related mortality rates in India, largely due to its unique plateau topography and the intense convection that builds up before and during the monsoon season. To address this, we developed a comprehensive machine learning pipeline that analyzes six years of daily lightning and meteorological data from 2017 to 2022 across all 22 districts. Our primary intention was to use these historical records to build a useful real-time forecasting tool for future lightning risk. These data from three separate sources have been integrated into a dataset comprising 48,180 district-day observations: from satellite LIS strike data with ground weather station measures like temperature, humidity, rainfall, pressure and wind.

We also integrated Convective Available Potential Energy (CAPE) from ERA5 reanalysis data to account for the atmospheric energy available to fuel thunderstorms. Given that lightning was recorded on only about 2.6% of district-days, a central focus of the project involved managing this significant class imbalance through feature engineering, including the use of lag features and cyclical month encodings.

After exploratory analysis and statistically testing which weather variables genuinely relate to lightning occurrence, a shared feature-engineering pipeline built lag features and cyclical month encodings. Two families of models were trained and compared on four different data slices (full dataset vs. the five highest-risk districts, and all-season vs. monsoon-only): an XGBoost classifier and a custom Hybrid Attention Convolutional-LSTM network that looks at the previous five days of weather to judge the risk of lightning on the next day. A separate, simpler XGBoost model — trained with SMOTE oversampling on single-day weather snapshots — was built for instant, one-shot predictions.

The top-performing setup (full dataset, all seasons) yielded a ROC-AUC score of 0.791 using the deep hybrid model, distinguishing the likelihood of lightning from its absence much better than a random baseline -raw accuracy figures, meanwhile, are misleading given the huge imbalance in lightning vs. No-lightning conditions. The trained models were hooked up to the free Open-Meteo weather API (which provides hourly current temp, humidity, rain, pressure, wind speed and CAPE for any location), wrapped in a Streamlit app and are there provided free of charge for anyone to specify a location/date to be instantly returned as Low/Medium/High likelihood for a given hour.

The rest of this post follows up on the how we gathered and cleaned the data, what our initial EDA and statistical analyses revealed, what feature engineering and modeling approaches we experimented with, what we found out as a result, and how we deployed the final solution – concluding with my opinion of its shortcomings and how we could improve on it in the future.

## **ACKNOWLEDGEMENT**

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I wish to acknowledge all who have extended their support and encouragement to me for completion of this project on “**Machine Learning Based Prediction and Real-Time Forecasting of Lightning Strikes in Jharkhand**”. I owe a huge debt of gratitude to my project guide whose help and supervision made each step clear. My project guide not only gave me immense inspiration, guidance and continuous advice but also a lot of technical expertise that helped me to overcome all obstacles that occurred during this work and resulted in significant betterment of the final product.

I also want to express my sincere gratitude to the teachers of the Department of Computer Science and Engineering, Birla Institute of Technology, Mesra for having supported me and created a helpful atmosphere to study and carry out this project.

I sincerely thank **DST PURSE Project 2022** and **Center for Advanced Research on Machine Learning for Geospatial Analytics and Solutions for Jharkhand State (ARMLEGS)** for allowing me to utilize their dataset, infrastructure and computing power.

I finally like to thank my parents and my friends for the patience, support and motivation they have provided me in all the stages of this work. My thanks and appreciation goes to each one who supported me, directly or indirectly for the accomplishment of this work.

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## Chapter 1: Introduction

### 1.1 Introduction

Lightning is the spontaneous emission of electric discharge occurring in conjunction with a thunderstorm where electric charge has accumulated within the cloud. This discharge can be within the same cloud, between clouds or between a cloud and the ground. When lightning occurs between a cloud and the ground, it is a strike that poses great threat and is dangerous for trees, people, houses and even power lines.

In a thunderstorm cloud, called a cumulus nimbus, upward and downward drafts of air carry along water particles, hail pellets and graupel through the storm cloud where opposite electrical charges collect at the top and bottom, causing a strike to be made when the air is unable to contain the charged regions. The loud sound created is called thunder.



*Fig 1.1 — Lightning over terrain*

The thousands of lightning strikes that occur around the world each year cause deaths, fires, destruction of communication and power infrastructure and affect air traffic. Weather forecasting for lightning has come a long way over the last decade and accuracy rates have increased significantly; however predicting the exact location and time of a lightning strike remains an issue, given that thunderstorms develop rapidly. The number of atmospheric factors that contribute to lightning strikes is complex, and just not one single variable (for example, temperature, humidity, pressure, precipitation, wind, instability). Temperature, moisture, pressure, precipitation, wind, and instability, which could be CAPE for instance, all affect lightning; however CAPE in isolation doesn't completely determine lightning.

Machine learning is a good fit for this problem because it is suitable in capturing non-linear relationships and interactions among numerous variables which are complex and would be otherwise overlooked. Thus this project was motivated to use a machine learning based system for lightning strike prediction in the districts of Jharkhand based on lightning strikes, historical weather parameters and CAPE data.

## 1.2 Lightning in India and Jharkhand

In terms of yearly thunderstorms and related deaths and damage to life, the country faces a huge toll. Many thunderstorms can be traced to its tropical climatic zone and monsoon-led weather. Jharkhand experiences many of them.

The terrain of Chota Nagpur plateau (Jharkhand), dominated by mountains and forest cover, is also an apt location for generating storms.

From April to September, Jharkhand has many lightning events every year. In addition, as most people here are dependent on farming and operate in the open, it is very susceptible to the vagaries of lightning and its consequences for agriculture, livestock, electricity lines, and communication facilities. Even though the weather department issues warnings, prediction at the district level, using machine learning models on data from this new branch of weather forecast, can provide great help to the government for disaster management.

## 1.3 Need for Lightning Prediction

Lightning is highly spontaneous and relies on the interplay of several atmospheric variables simultaneously so, static threshold criteria is not enough. It's a rare natural disaster relative to normal weather events which poses forecasting challenges. With more recent availability of and accessible costs of large amounts of data on weather and satellite imagery, data-driven models to forecast natural disasters may be viable. This will give the ability for governments, farmers, and other interested parties to take precautions to minimize deaths and destruction prior to the occurrence.

## 1.4 Problem Statement

Here, the dataset has a very severe class imbalance with only a minority class with lightning, and from multiple disparate sources that required significant data wrangling to make use of the same. The idea was to build a system that uses satellite lightning data along with weather records and CAPE that would then forecast occurrence of lightning by district per day in Jharkhand, making real-time predictions available via a web application using free weather forecast sources.

## 1.5 Objectives

- Collect and merge lightning, weather, and CAPE datasets from multiple sources.
- Clean and preprocess the merged data.
- Explore spatial and seasonal lightning patterns across Jharkhand.
- Test which weather variables statistically relate to lightning occurrence.
- Build and compare machine learning models for district-level prediction.
- Deploy the trained model as a Streamlit app using live Open-Meteo data.

## 1.6 Scope of the Work

We present the model to predict the daily occurrence of lightning at the district level in Jharkhand using data from 2017-2022 with LIS satellite data, ground station data and ERA5 CAPE as features.

This study is conducted to demonstrate what can be achieved by utilizing open-access data and open-source tools. We aim only to forecast the probability of the lightning occurrence at district level and not to predict precise lightning strikes or to conduct now-casting at minute level.

## **1.7 Organization of the Report**

Chapter 2 Related work of lightning prediction, 3 Materials and methods (including Data, Pre-processing, Feature Engineering, Models, Deployment), 4 Results, 5 Conclusions and discussions (including Lightning risk assessment per district, Limitations and future work).

## Chapter 2: Literature Review

### 2.1 The Physical Basis of Lightning: Why CAPE Matters

In general, most studies of weather and precipitation have relied on just a few key parameters about atmospheric instability. CAPE, which is known as the **Convective Available Potential Energy** is likely the most well-known among these. It represents how much energy would be added to a warm bubble of moist air that is at the surface if that bubble continued to rise.

So high CAPE is a sign that the atmosphere is very ready for big strong tall storm clouds and therefore very prone to those types of clouds with internal updrafts and charge separation that occurs with lightning.

Humidity temperature falling pressure are all associated with thunderstorms as well and tell something about the amount of moisture present and an incoming low pressure system.

### 2.2 Related Work

I found an excellent guide to help me, a 2023 paper in the International Journal of Climatology by Chatterjee et al., that split India into seven lightning climatology zones based on Cape, humidity, temperature, and the K-index, and also compares models - Random Forest was the winner there. Even more useful for my project is Mandal, Chatterjee, and Das's 2024 paper predicting lightning density for North-East India using the exact same ERA5 and LIS variables I'm using and SHAP to determine the most important lagged monthly variables. This is functionally the same feature importance task as I'm performing on a daily level.

My mind was also snagged by an early 2025 paper in Meteorology and Atmospheric Physics creating a "Thunderstorm Potential Lightning Index" for Guwahati which states that the 'tall and fat' CAPE profile results in much higher lightning frequency and probability compared to 'tall and skinny' profile despite the latter being a more frequent occurrence.

That the shape of the profile matters - as much as the sheer magnitude - came into consideration when formulating my variables. Beyond that, broader review papers (Wang et al., 2025; Zhang et al., 2025) on the field seem to indicate a trend away from simple threshold-based nowcasting to primarily two major directions: tree based ensembles (Random Forest, Gradient Boosting, XGBoost) and deep sequence models (LSTM, CNNs etc. ), something that the two primary architectures chosen for this study mirror exactly, a selection which would, it is hoped, make for a fair comparison.

## Chapter 3: Methodology

### 3.1 Data Sources and Study Area

All 22 districts of Jharkhand are covered here. I pulled together four different sources, listed below.

| Source                         | File                                                     | What it provides                                                                            |
|--------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Lightning Strikes (LIS)        | All District Lightning Data 2017-2022                    | Satellite- based Lightning Sensor strike events and counts, per district per day            |
| Ground weather (MET)           | Jharkhand ALL-District Weather data                      | Daily Pressure (hPa), Relative Humidity(%), Rainfall (mm), Temperature(°C), Wind Speed      |
| Atmospheric Instability (CAPE) | ERA5 reanalysis NetCDF extracts (3 zip files, 2017-2022) | Convective Available Potential Energy, a measure of storm-fuelling energy in the atmosphere |
| Geography                      | Jharkhand district shapefile                             | District boundaries and centroids, used for maps and for fetching per-district elevation    |

Table 3.1 — Data sources used in the project

Two data-quality issues arose during the loading process. First, some of the “.xls” files actually contained plain CSV text with an incorrect extension. This often arises from exporting data from Google Earth Engine, so the loader now attempts to detect and fall back to a CSV reader. Second, by chance, all three of the CAPE zip archives contained a NetCDF file with the identical internal name.

The previous extraction code blindly extracted each year’s file into the same working directory, meaning the prior year was always overwritten. Resultantly, only the last two years of CAPE ever ended up on disk, leading to the loader often returning a CAPE value of 0 – including on most actual lightning days. After ensuring that each zip file was extracted to its own unique directory prior to processing, the proportion of rows with a valid, non-zero CAPE value jumped from something small to over 91%!

### 3.2 Data Cleaning and Merging

These three sources were combined by District and Date. Lightning events were rolled up to the district-day, and weather data plus CAPE was joined by district-day. `Lightning_Occurred` is a binary indicator, 1 if any lightning strikes were detected in the district on that day, and 0 otherwise.

We are left with a table with 48,180 rows, and 14 columns for the years January 1, 2017 to December 31, 2022 for all 22 districts. There are no more missing values.

### 3.3 Exploratory Data Analysis

Before modelling, the merged data was checked visually for which districts, seasons, and years see the most lightning, and how the six weather variables relate to each other. Figure 3.1 shows the spatial spread of the top five lightning-prone districts — the convex hull and bounding box of every

strike recorded in each one. These findings, along with the district and seasonal breakdowns and the correlation matrix, are presented with their charts in Sections 4.1 and 4.2.

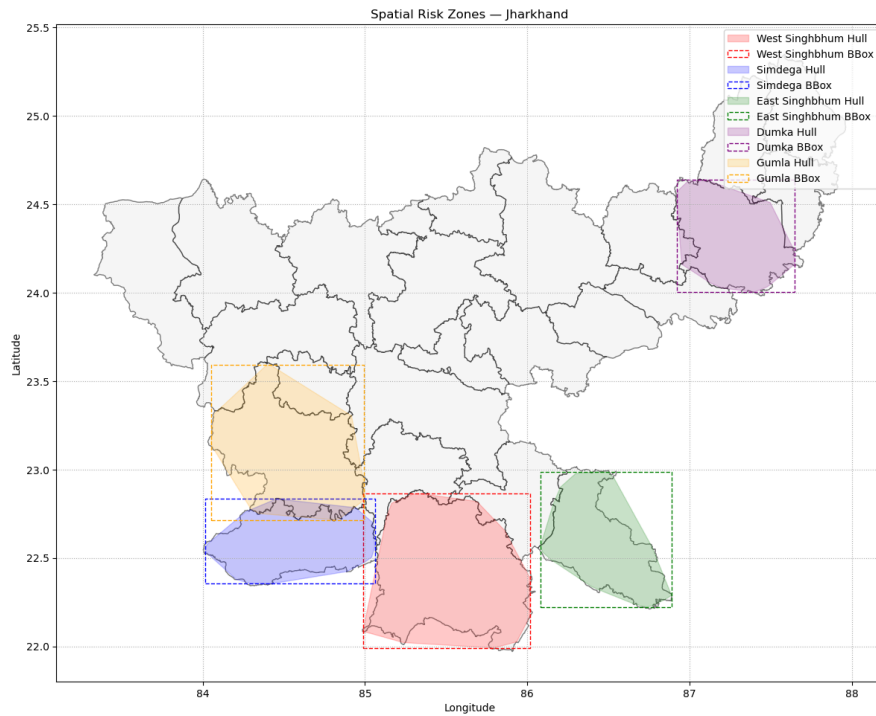


Fig 3.1 — Spatial risk zones of the top 5 lightning-prone districts of Jharkhand

### 3.4 Statistical Significance Testing

Just by looking at the charts won't tell you whether something's a pattern or just a coincidence, so I used two different statistical tests for each of the six variables versus `Lightning_Occurred`; A point-biserial correlation, which measures how well the two are related to each other and in what direction, and a Mann-Whitney U test, which helps determine whether lightning-containing days are truly different from non-lightning days. I won't go into the details of the full report but the answer to both tests was that for every variable, at a significance level of 5%, there is a statistically significant relationship between lightning and the variables. CAPE proved to be the most statistically significant.

### 3.5 Feature Engineering

The same shared feature function was used on every model so results remain comparable between slices of data. It begins with the six raw weather features and adds two lag features for each (yesterday's and the day before's reading so the model knows if weather conditions are building or improving) and a cyclical sine and cosine encoding for the month (so Dec and Jan aren't interpreted as miles apart).

Feature group contains Same day reading, Lag features, Cyclical month encoding with count as 6, 12, 2 respectively. Talking about the description of these counts:

- Same day reading - Pressure, humidity, rainfall, temperature, wind, CAPE
- Lag features - Same 6 variables, 1 and 2 days earlier
- Cyclic month encoding - sin/cos of calendar month

Additional physically-based features were subsequently included to this layer at the request of my team lead, including: dew point and dew-point depression, humidity-weighted CAPE index, IMD style rain intensity bucket, daily pressure tendency and various interaction features (CAPE Humidity, Temperature Humidity, Rainfall Wind Speed), that feed the additional analysis performed in section 4.7.

## 3.6 Model Architectures

### 3.6.1 Sequence Models (5-Day Look-Back)

Now For The Core Problem – Is It Going to Storm Tomorrow Based On The Weather in The Past 5 Days. Each district was converted into a sequence of 5 day overlapping sequences. On these we trained two models: XGBoost: Each 5 day sequence is flattened into a single long row with 100 columns (5 days \* 20 features), because tree based models are blind to time ordering.

To overcome the class imbalance, a `scale_pos_weight` parameter was passed into XGBoost which calculated the ratio of rows with non-storming to rows with storming events.

Hybrid Attention CNN-LSTM: A neural network trained in TensorFlow/Keras, which does take time into account. A first layer with a 1D convolutional kernel learns some basic local time correlations. Then these short sequences are passed into a Bidirectional LSTM, which traces the evolution of the local patterns across the 5 day window. An attention layer helps weigh the contributions from individual days more appropriately, before a final dense layer outputs the probabilities.

Both used a simple 80/20 split to train/test (and 15% of the train was reserved for a validation set).

Feature scaling was performed with a RobustScaler fit only to the training data. The decision boundary for the deep model was optimized on the validation data by maximizing F1 score, then the model with this decision boundary was applied to the held-out test set.

### 3.6.2 Manual / Live Single-Snapshot Model

I trained a simpler model XGBoost with one day of six weather measurements without lags – an app doesn't have the luxury of 5 days of past information available for a new location on the fly. To even the classes out from around 850 lightning days vs. 31,912 days no lightning on my split I used SMOTE, balancing it so we got 31,912 lightning vs. 31,912 no lightning, only applied to the training section. The optimal threshold using the same technique as the deep model turned out to be 0.7785. I also added a safety catch to the app; if you put in an impossible value it bounds the result to the edges of the training value range and warns you it was a wild guess.

## 3.7 Evaluation Metrics

Lightning happens 2.6% of the time, so plain accuracy isn't really an option as that gets us ~97% just by saying "no" to lightning every single day. Instead I decided to measure accuracy on precision, recall for the minority (lightning) class, F1-score, the confusion matrix and ROC-AUC. Accuracy is still reported, but purely for reference. ROC-AUC in particular measures how well a model separates the two classes across every possible threshold, regardless of the imbalance — 0.5 means no better than a coin flip, 1.0 means perfect separation.

### **3.8 Deployment Architecture: Streamlit and the Open-Meteo API**

Since it's difficult to understand through a notebook, I merged it with Open-Meteeo, a free weather API site. It directly fetches the meteorological data and provides us without using any API key. I can retrieve the current hourly temperature, humidity, rain, pressure etc. for any coordinate. This helps us a lot in showing our project into real world scenarios which will help us in getting live lightning risk output.

All of this, including the trained model, its fitted scaler, its threshold found during optimization, and the real-time data retrieval logic lives within a Streamlit app, which on a location selection (search or browse), will retrieve the current conditions there, scales it like the training data, and feeds it through the model to output a Low/Medium/High lightning risk. It's also possible to query historical dates in live data, and when too distant for live data, it enters into a historical/climatological mode (clearly labeling when this happens).



## Chapter 4: Results and Discussion

### 4.1 Dataset Overview and Class Imbalance

To get a sense of what we were working with, we looked at the full dataset of 48,180 district-day records. So as it appears lightning days were actually few and far between – only 1250 days (2.59% of the total) witnessed a strike with 46930 (97.41%) remaining placid, translating to approximately one lightning day for every 37.5 quiet days. The consistency in these values was actually fascinating – from 2.4% in 2018 to 3.0% in 2020 the rate never varied significantly and appears to be a fixture of weather in Jharkhand, rather than an anomaly.

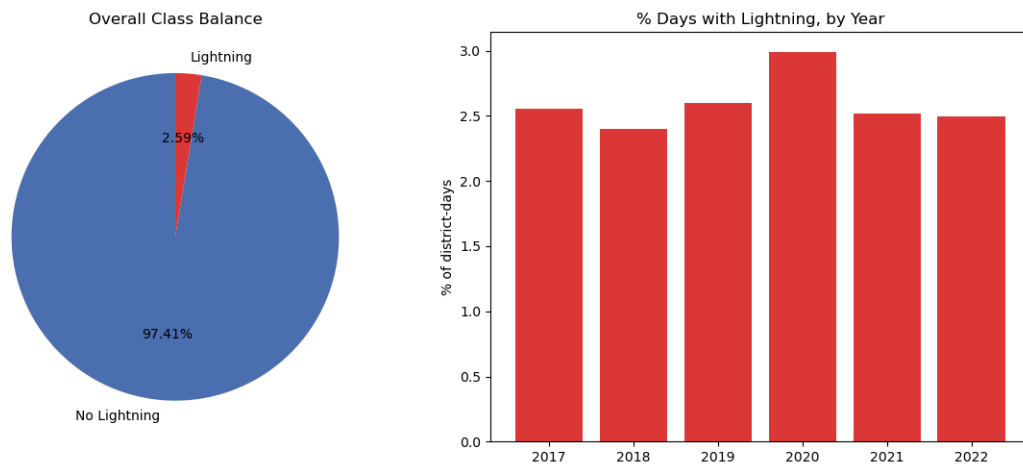


Fig 4.1 — Overall Class balance and yearly lightning rate(2017-2022)

### 4.2 Spatial and Seasonal Patterns

When they plotted all lightning strikes, the intensity was not evenly distributed, as lightning strikes were concentrated in several pockets. West Singhbhum has emerged as the state's major hotspot, recording 36,675 incidents in six years. The district has 1.6 times more lightning strikes than Simdega (23,232), the state's second-highest hotbed for the natural phenomenon. The other three districts among top five active regions include Dumka, Gumla and East Singhbhum..

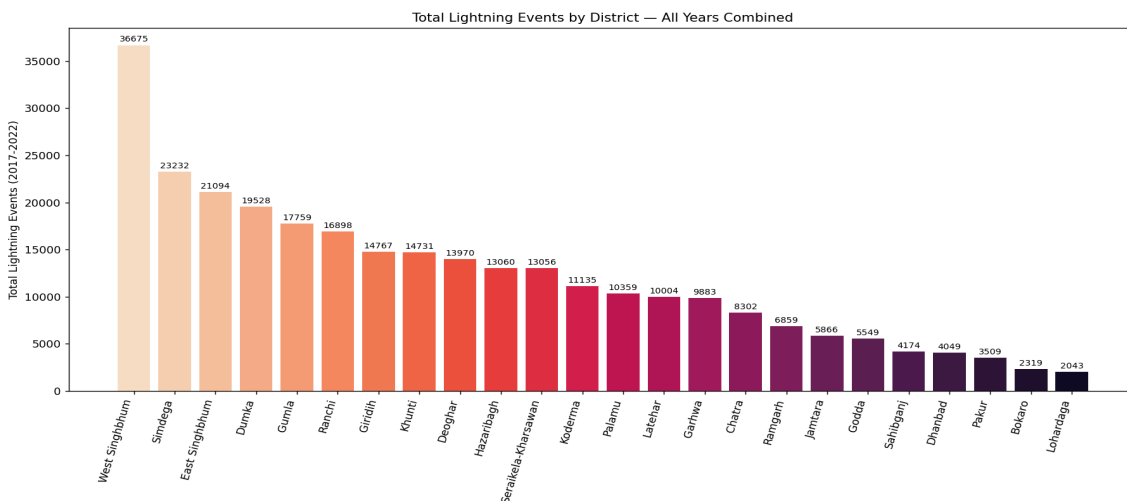


Fig 4.2 — Total lightning events by district, all districts combined (2017–2022)

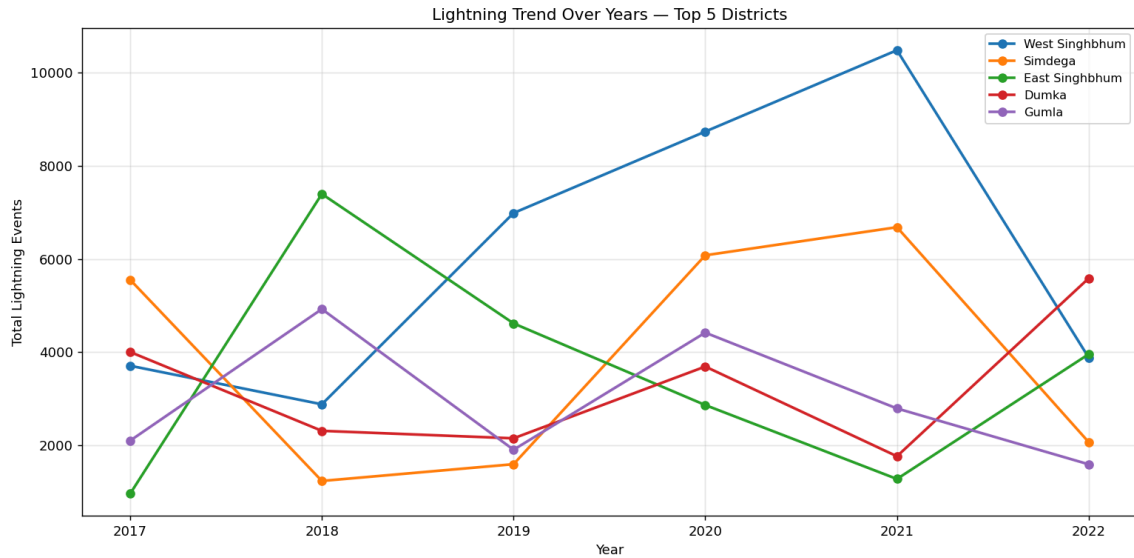


Fig 4.3 — Lightning trend over years, top 5 districts

We also compared how often lightning happens (frequency) against how many strikes occur when it does (intensity). The most frequent and most severe region is uniquely West Singhbhum. Most of the other districts are pretty much the same line on the diagonal: frequency of lightning in those areas has a strong, predictable correlation to total lightning strikes. So our two measurements are being interpreted the same way.

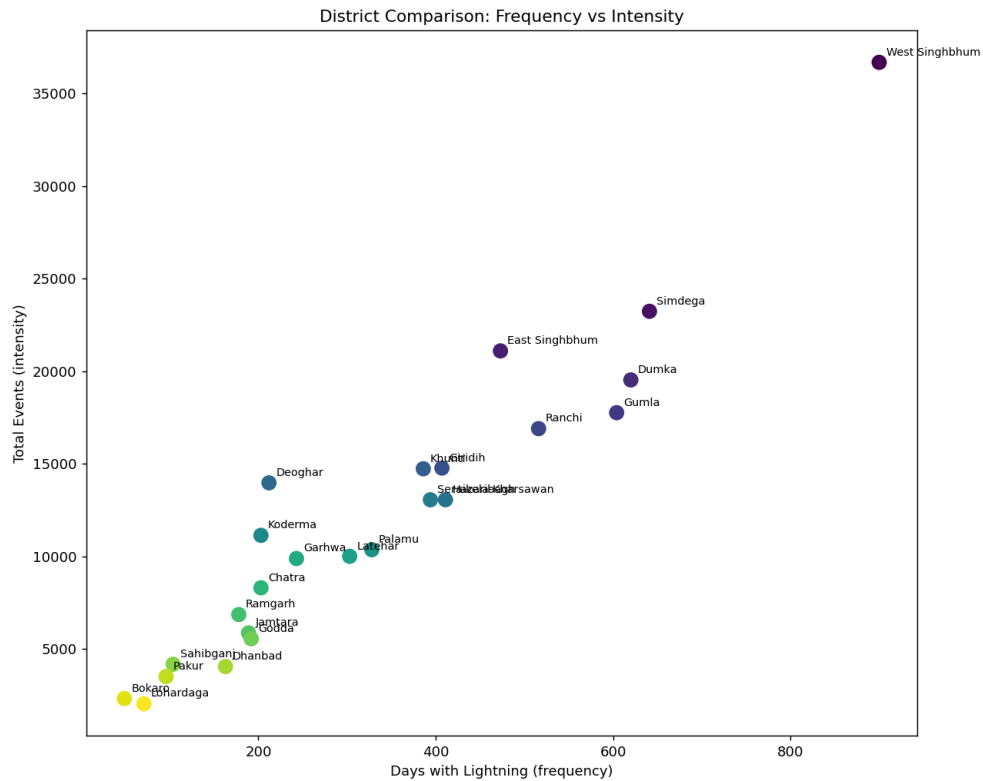
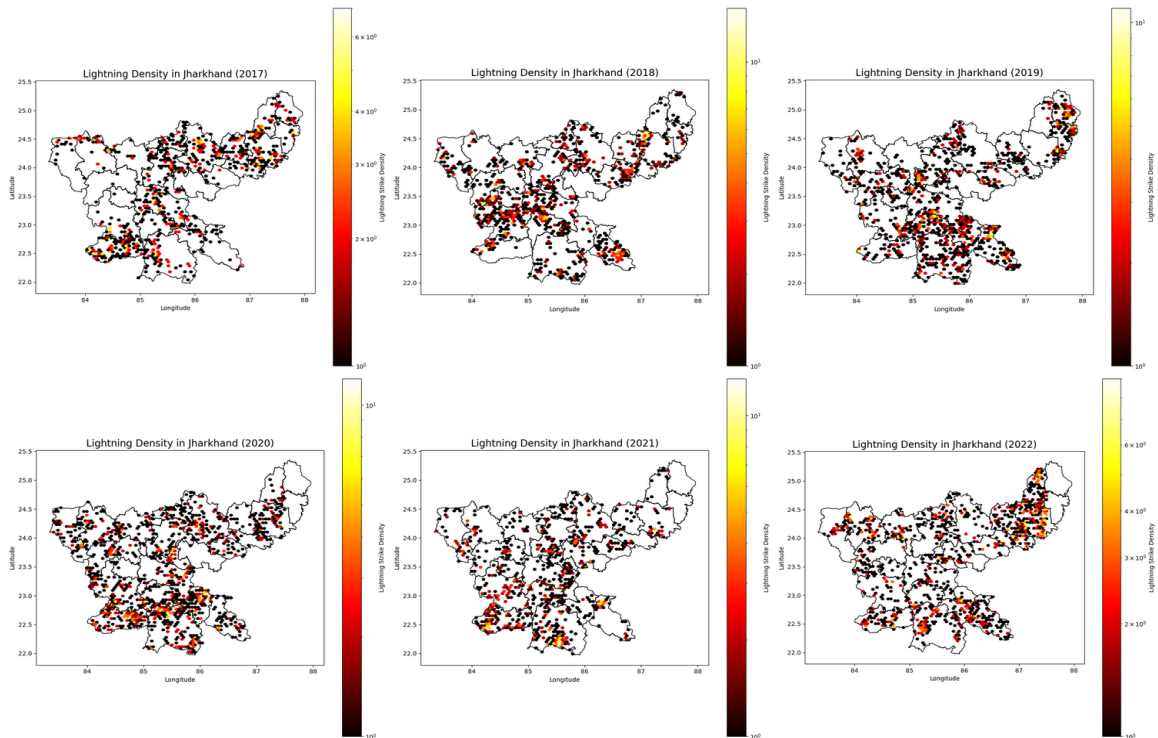


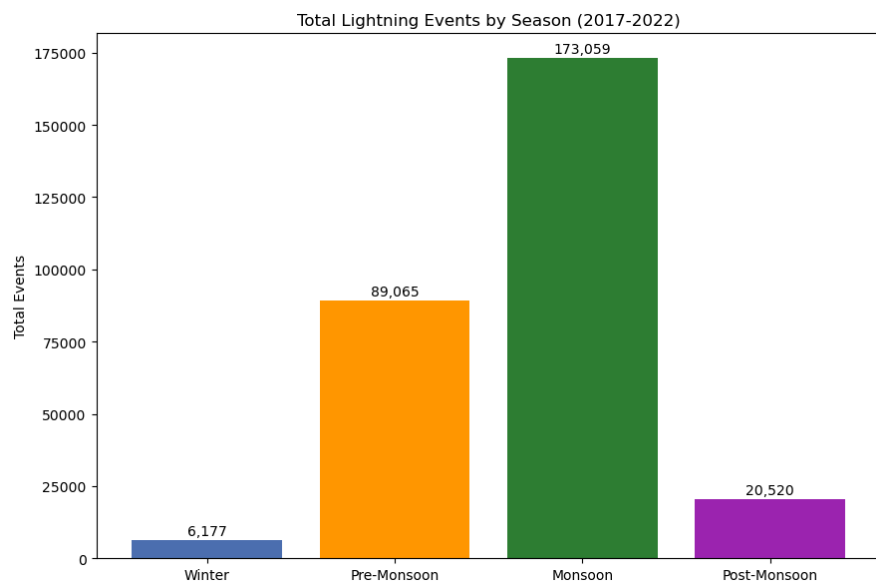
Fig 4.4 — District comparison: frequency (days with lightning) vs. intensity (total events)

We put together year-by-year density maps to check this wasn't just some long term average over the data. Plotting strike coordinates from the raw year-by-year density plots, the “hotspot” for West Singhbhum, Simdega and East Singhbhum clearly emerges as the densest cluster for every single year.



*Fig 4.5 — Lightning density across Jharkhand, year by year (2017–2022)*

Timing is also important like location. Most of the lightning activity takes place during the monsoon season from June to September that accounts for nearly 60% of the entire activity. Although, the pre-monsoon period (March-May) also contributes significantly to the process, it's eerily silent in the winter period (December-February).



*Fig 4.6 — Total lightning events by season (2017–2022)*

We even wondered if lightning was becoming more common over time. Our trend test showed a slight upward slope — about 2,576 extra events per year — but with a p-value of 0.253, it wasn't statistically significant. With only six years of data, we can't say for sure if there's a long-term increase or if the big spike we saw in 2020 was just natural year-to-year variation.

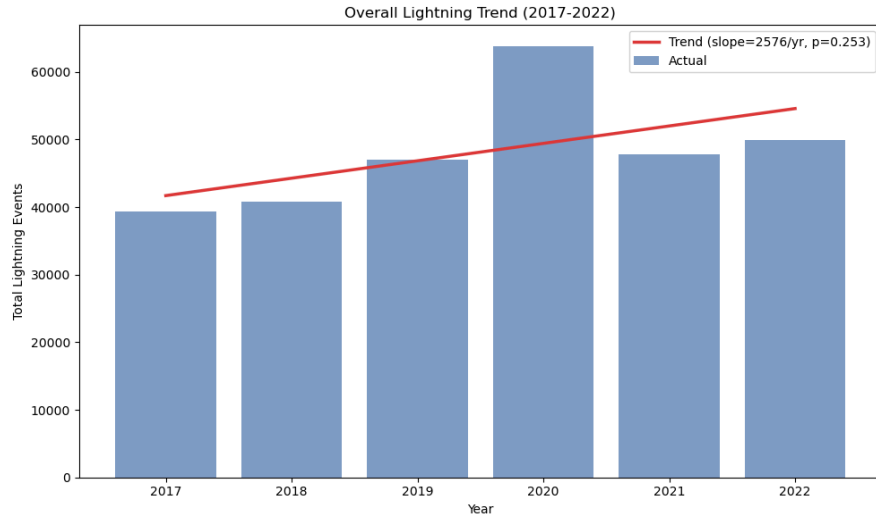


Fig 4.7 — Yearly lightning trend with linear regression fit

Temperature and CAPE correlate the strongest out of all the pairs of six weather variables, ( $r = 0.61$ ) and that is intuitively obvious: the higher the temperature of the surface air the greater amount of energy will there be (measured by CAPE) capable of creating the buoyant forces. The relationship between humidity and rainfall also has moderate correlation ( $r = 0.50$ ). None of the variables correlates with the plain event count individually quite strongly, that was anticipated: the conditions for lightning development are multiple and therefore all the models were based on all six variables and their respective lags rather than just one.

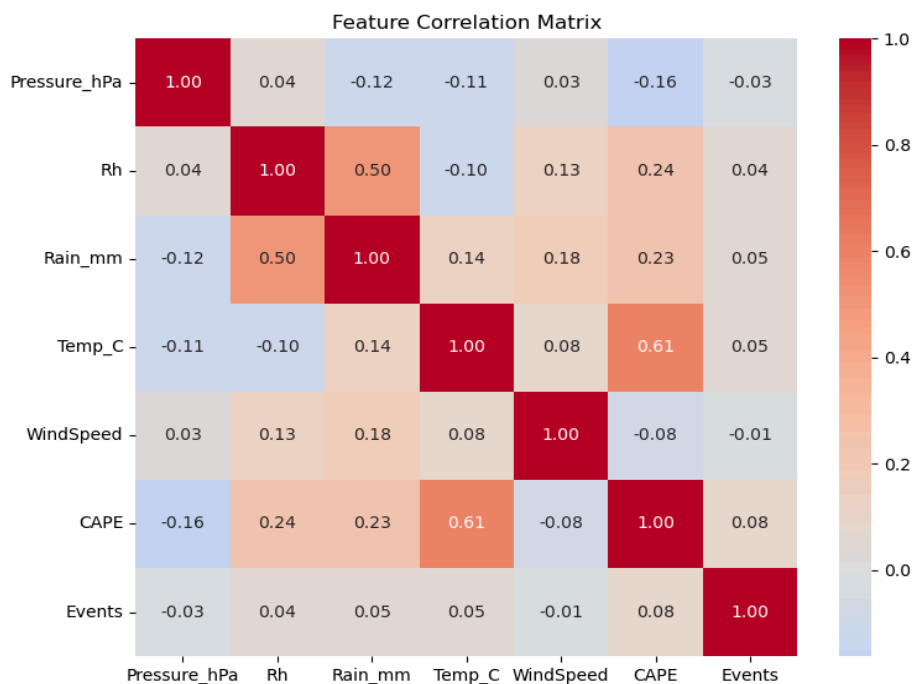


Fig 4.8 — Correlation matrix of the six weather features

### 4.3 Statistical Significance Results

Apart from just charts, we ran formal statistical tests. Table 4.1 shows that every single one of our six weather variables had a significant relationship with lightning (all p-values were well below 0.05). Unsurprisingly, Cape performed the strongest out of the whole group, as expected from the existing literature we'd read above.

| Feature       | Correlation (r) | p-value | Direction                           |
|---------------|-----------------|---------|-------------------------------------|
| CAPE          | 0.1261          | <0.001  | Higher CAPE→more lightning          |
| Rain_mm       | 0.0856          | <0.001  | Higher rainfall → more lightning    |
| Temp_C        | 0.0806          | <0.001  | Higher temperature → more lightning |
| Rh (Humidity) | 0.0625          | <0.001  | Higher humidity → more lightning    |
| Pressure_hPa  | -0.0564         | <0.001  | Lower pressure → more lightning     |
| WindSpeed     | -0.0168         | <0.001  | Lower wind speed → more lightning   |

Table 4.1 — Statistical significance of weather features vs. lightning

### 4.4 Model Performance Comparison

To gauge just how sensitive this problem is to the geographical and seasonal span of the data, I ran four training scenarios: the full dataset over all seasons, the top five districts only, the full dataset across the months of monsoon and the top five districts during the months of monsoon. While raw accuracy figures look almost perfect, that is, thanks to the large class imbalance.

| Run                                   | Rows   | Districts | XGBoost Acc. | Hybrid Acc. | Hybrid ROC-AUC |
|---------------------------------------|--------|-----------|--------------|-------------|----------------|
| Full Data — All Districts, All Season | 48,180 | 22        | 95.39%       | 91.29%      | 0.791          |
| Top 5 Districts — All Season          | 8,760  | 5         | 96.17%       | 91.13%      | 0.766          |
| Full Data — Monsoon Only              | 16,104 | 22        | 94.78%       | 94.03%      | 0.772          |
| Top 5 Districts — Monsoon Only        | 2,928  | 5         | 92.78%       | 89.69%      | 0.639          |

Table 4.2 — Model comparison across the four training runs

An XGBoost run on the entire dataset over all seasons attained an overall accuracy of 95.39%, but had a poor precision of 0.19 and a recall of 0.23 for the less frequent “lightning” class. The figure we care about, the ROC-AUC, proves this point, with the full dataset over all seasons proving to be best

hybrid (0.791) among all four training regimes and the smallest subset (top 5 districts, monsoon only) being the worst (0.648), likely as a consequence of only 2,928 samples available to train the deep network on.

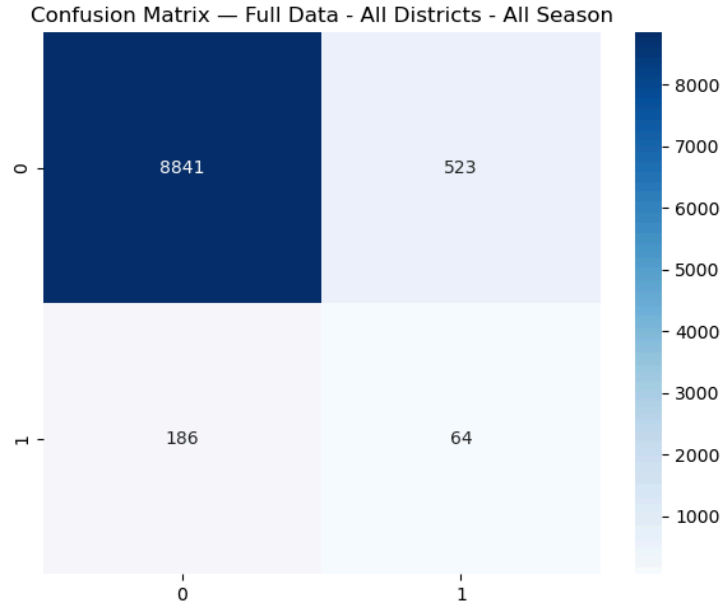


Fig 4.9 — Confusion matrix — Full Data, All Districts, All Season (Hybrid model)

This highlights the real trade-off of our confusion matrix: of 9,364 non-lightning days, we predicted 8,841 correctly (which includes 523 false alarms). Of 250 actual lightning days, we predicted just 64 of them correctly (and missed 186 of them). This is by design, for a low frequency and high impact event, where we are trading off precision for recall, instead of optimising for overall accuracy - because the penalty for missing a day of lightning greatly exceeds the cost of an unnecessary alert. However, missing 186 out of 250 positive predictions is a clear illustration of the class-imbalance limitation we discuss in chapter 5 - we could surely increase our recall here, probably at the price of some more false alarms.

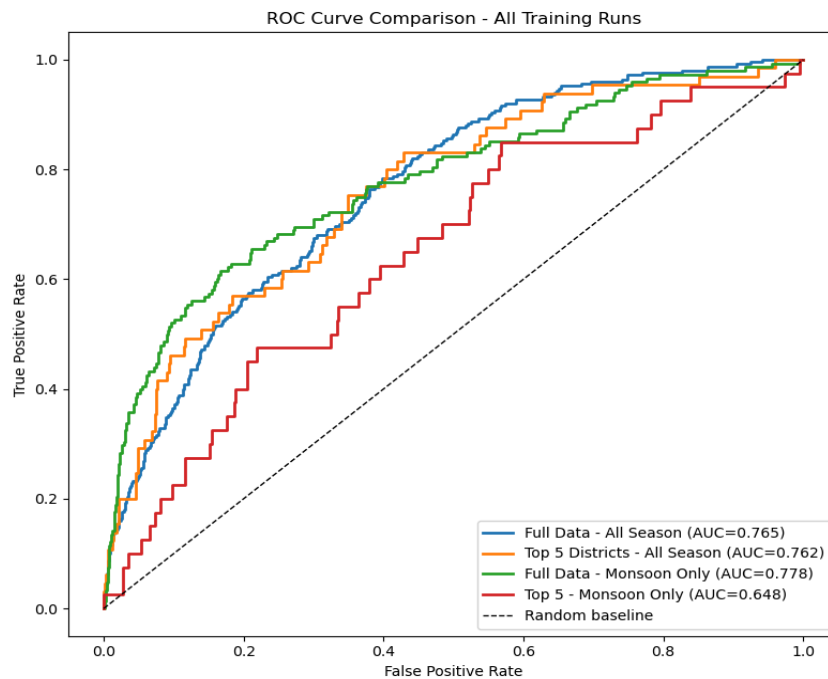


Fig 4.10 — ROC curve comparison, all four training runs

## 4.5 Manual and Live Prediction Tool Behaviour

This manual single-snapshot model was sanity checked using ten random real lightning days. The model gave quite high probability (between 0.03 to 0.95 with most  $>0.5$ ) on conditions that were actual lightning days – a good sign that it actually learned physical dependencies, not noise. The feature importance ranking put rain fall (26.5%) and CAPE (23.7%) on the top 2 positions with the other inputs following at the order of importance such as pressure (17.7%), wind speed (10.9%), humidity (10.7%) and temperature (10.5%), roughly in line with that of Section 4.3.

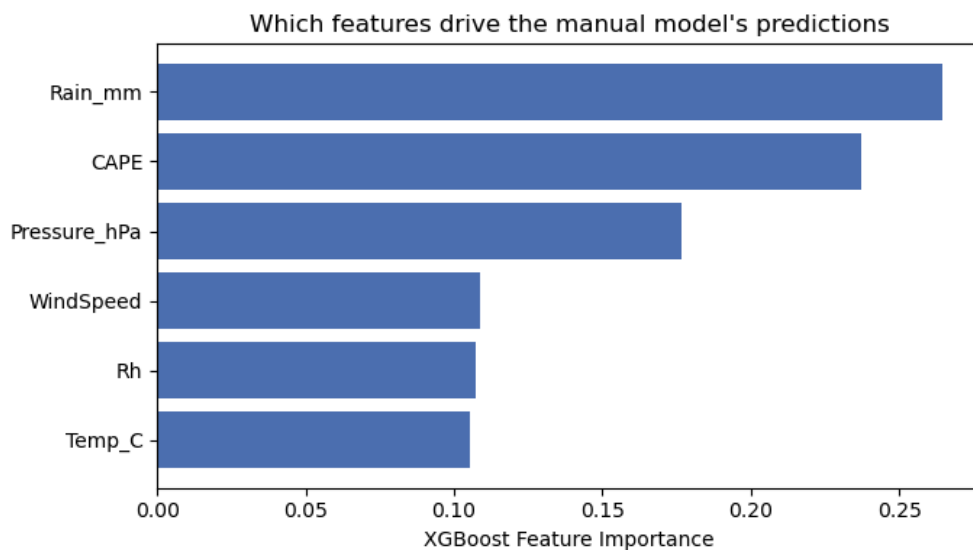


Fig 4.11 — XGBoost feature importance for the manual single-snapshot model

As an example the model predicted 0.83% with an output of “Low risk” in relation to a hot humid day with the following conditions: Temperature of 30C, relative humidity of 78%, amount of rain of 50mm, pressure of 986hPa, wind speed of 1m/s, and CAPE value of 500. This was below the set threshold value of 0.7785%. This makes sense physically; usually, very heavy rain paired with low CAPE means a storm has already spent its energy rather than building up for a new strike.

## 4.6 Forecasting and Risk Lookup

On top of the trained hybrid model are two convenience tools. A rolling next-N-days outlook extends the probability forward for a particular district by extending its latest known weather pattern, a climatology-driven view and not a real numerical weather forecast as no one can predict future weather yet. A date-and-district risk lookup can also be used by the user to inquire about a specific date. If this date falls within the bounds of past data for that district the actual observed weather for that district will be used. If that date lies beyond past observed data, standard climate data for the time of year will be used. The view will state on which basis it was created.

The images below show the real running application with actual live weather data from Open-Meteo on the time of the inquiry. A single district view provides relative-risk percentile, Low/Medium/High category, and actual weather on the backend of the probability estimate, and the map view displays

each district as a coloured symbol, with the same data available upon selection of a district on the map.

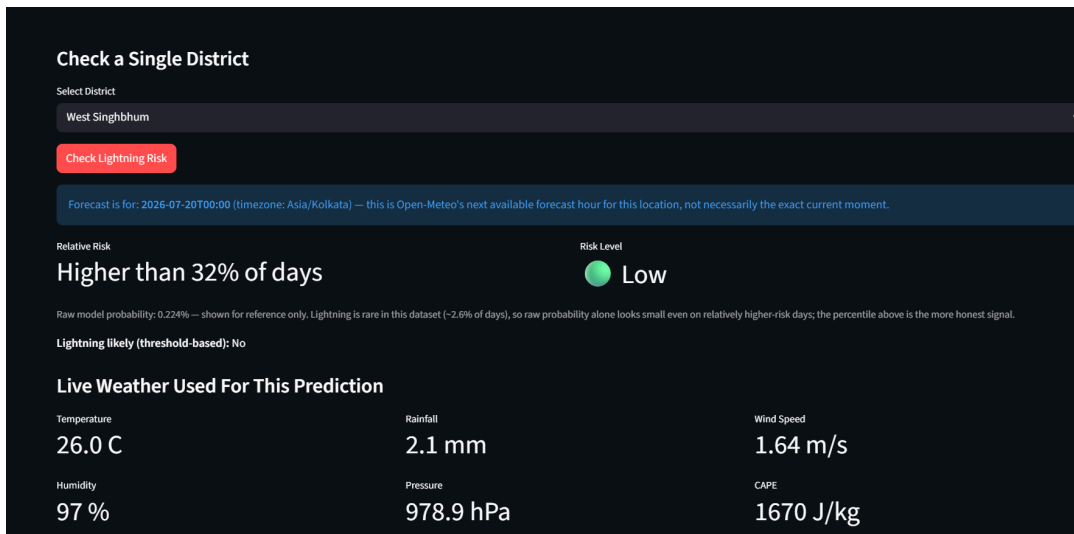


Fig 4.12a — Deployed app: single-district risk check (West Singhbhum, live data)

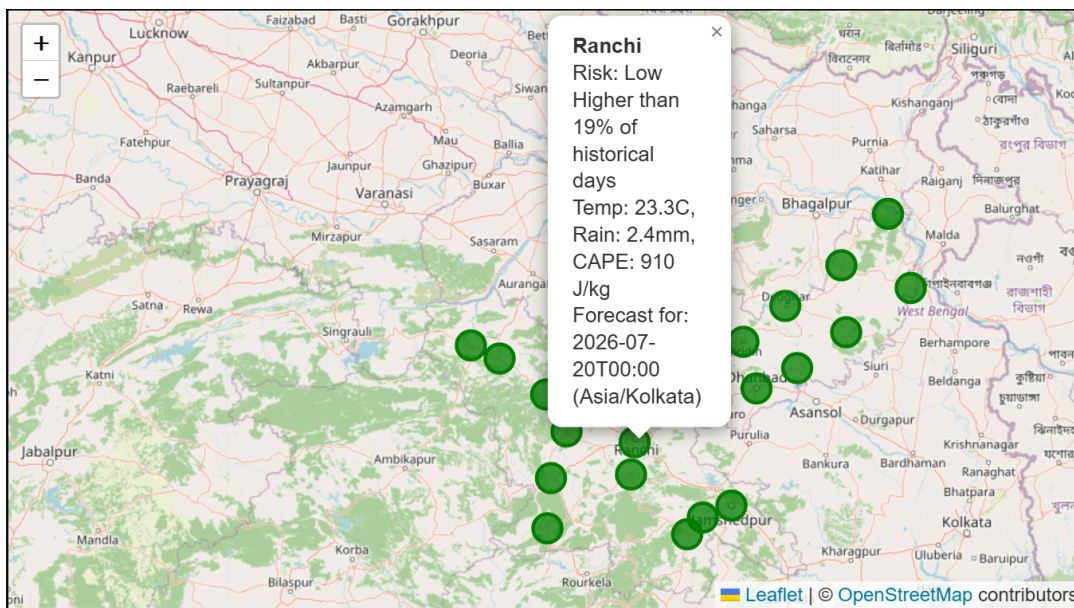


Fig 4.12b — Deployed app: statewide map view with live per-district risk popups

## 4.7 Additional Analysis

After our initial project review, we decided to dig a bit deeper and expand our analysis in a few new directions.

We checked if a district's elevation had a direct impact on lightning frequency, but the correlation was weak and not statistically significant ( $r = 0.359$ ,  $p = 0.101$ ). Looks like the height above ground is far less important than something like CAPE and temperature in Jharkhand.

We also ran a Principal Component Analysis (PCA) on our extra features. We found that just 5 of the 8 new variables explained over 91% of the variance. This tells us the new features we engineered are actually adding new, useful information rather than just repeating what was already there.



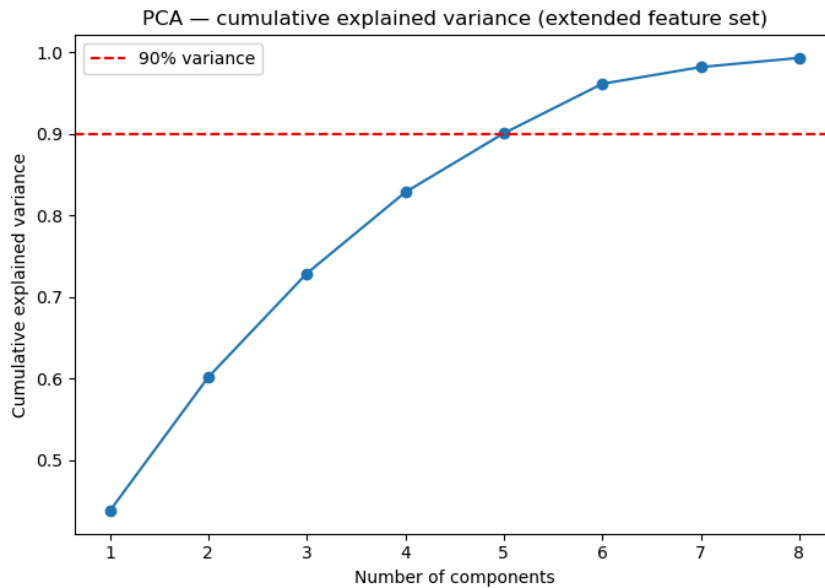


Fig 4.13 — PCA cumulative explained variance (extended feature set)

This confirmed via bootstrap logistic regression that temperature, pressure, CAPE and humidity are all significant predictors. We then ran a logistic regression model controlling for these variables and found that the raw rainfall and the daily changes in pressure then became non-significant, and were effectively being pulled along for the ride by the stronger factors.

Finally we run 6 different classical models over the same balanced data and compare their performance in the table below (Table 4.3).

| Model               | Accuracy | ROC-AUC |
|---------------------|----------|---------|
| Gradient Boosting   | 69.4%    | 0.785   |
| Random Forest       | 93.6%    | 0.784   |
| XGBoost             | 83.7%    | 0.773   |
| Logistic Regression | 62.4%    | 0.752   |

Table 4.3 — Extended model comparison on SMOTE-balanced data

In the end, the four models finished up in a very similar performance zone (ROC-AUC between 0.752 and 0.785). We even tried SVM (0.774) and KNN (0.710) for reference. What's more, it was also clearly evident that for this data type, features and handling class imbalance is a significantly more important factor than picking a model.

### Lightning Strike Prediction — Jharkhand

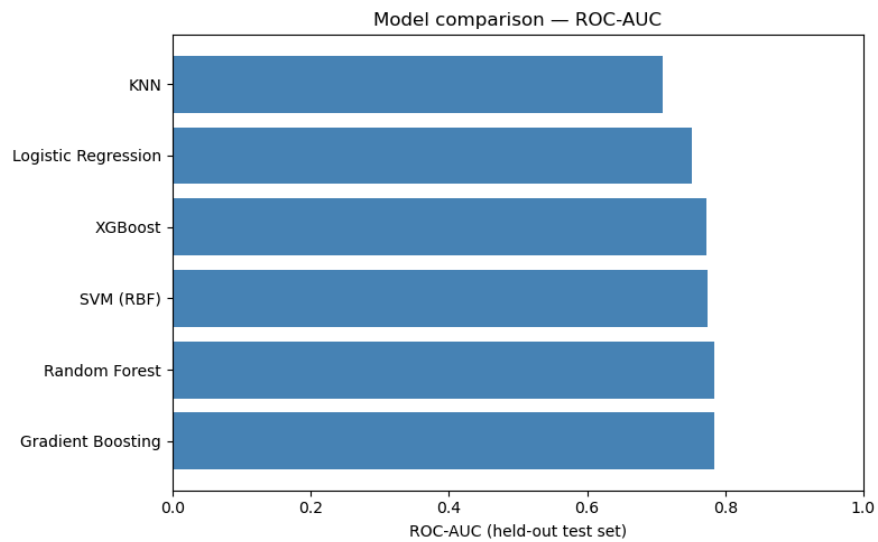


Fig 4.14 — Model comparison, ROC-AUC across six algorithms tested on SMOTE-balanced data

We've also built a choropleth map that displays average anticipated lightning risk for each district assuming standard weather conditions. This can be a useful visual at a glance at regions which may be naturally more likely to experience lightning, independent of what the weather looks like today as the app can inform you of.

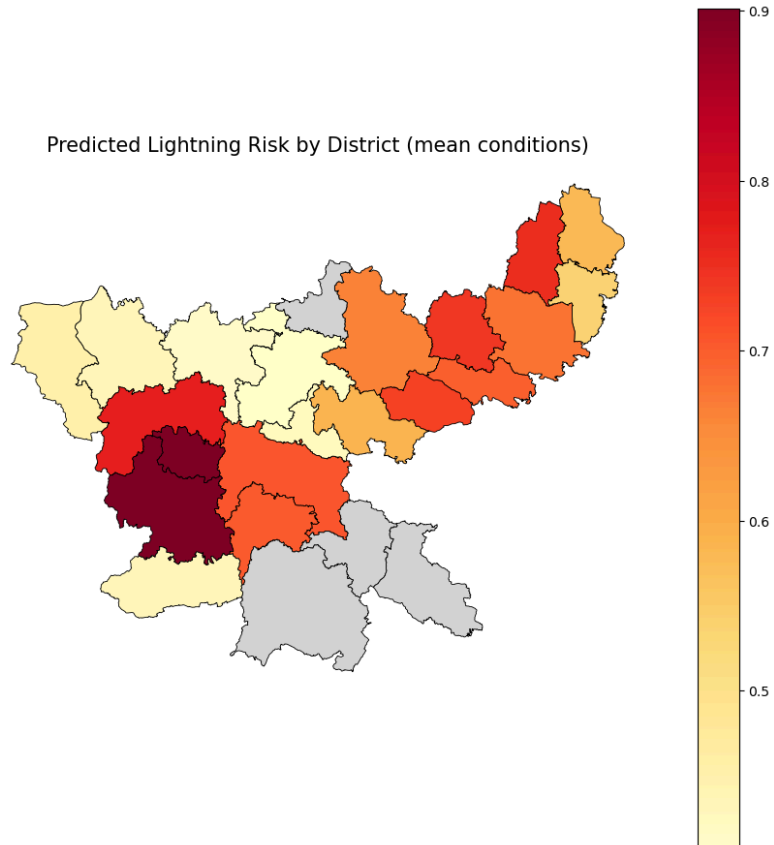


Fig 4.15 — District-wise predicted lightning risk (mean local conditions)

## Chapter 5: Conclusion

### 5.1 Summary of Findings

In this project, we converted an unstructured and noisy collection of raw data to a functional predictor that outputs a lightning strike probability for anywhere in Jharkhand. After processing 6 years of satellite and weather data, we confirmed the top six predictors for strikes – including variables like CAPE, rainfall, and temperature – while also narrowing down our choices from more than two dozen! Our best model performed well, generating an ROC-AUC score of 0.791- an excellent result for the very challenging task of accurately predicting an event that occurs with such rarity. To make our findings more accessible, we deployed our predictions as an interactive Streamlit app that can query live weather conditions and output the local likelihood of a lightning strike.

### 5.2 Where to Focus: A District-Wise Guide

We've divided our 42 districts into three levels of alert over our six-year study. It's not meant to substitute a daily forecast, but gives a good basis for deciding where to deploy a new sensor or run a safety program.

| Alert Tier                    | Districts                                                                                           | Basis                           |
|-------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------|
| High — priority monitoring    | West Singhbhum, Simdega, East Singhbhum, Dumka, Gumla                                               | >17,000 total events, 2017–2022 |
| Moderate — regular monitoring | Ranchi, Giridih, Khunti, Deoghar, Hazaribagh, Seraikela-Kharsawan, Koderma, Palamu, Latehar, Garhwa | ~9,800–17,000 total events      |
| Lower — baseline awareness    | Chatra, Ramgarh, Jamtara, Godda, Sahibganj, Dhanbad, Pakur, Bokaro, Lohardaga                       | <9,000 total events             |

Table 5.1 — District-wise alert tiers, for future monitoring

The “High” tier districts receive a very significant percentage of the state’s lightning strikes, hence these would be the highest in terms of priority for monitoring. “Moderate” tier also receives relatively significant activity, especially in the months of monsoon. “Lower” tier received a lower number of lightning strikes in the state, still with more than 2,000 in Bokaro and Lohardaga, these can not be considered risk free. They are simply the last priority for our resources.

### 5.3 Limitations

- Class imbalance is still the primary problem. Class weights and SMOTE address some of that, but recall and precision on the lightning class are still orders of magnitude lower than the reported headline accuracy – some lightning days are still missed entirely.
- The rolling forecast uses persistence and climatology not a numerical weather model. Realistically there's no way of knowing weather that far into the future, so it's a prediction, not a forecast in any formal sense.
- There's enough data (6 years) to establish seasonal trends but not enough to make statements about genuine long-term trends.

- The elevation/lightning relationship in Section 4.7 uses only 22 data points at district level, so a larger data set might improve confidence in the result.
- The live tool depends entirely on the accuracy and reliability of Open-Meteo's predictions.

## 5.4 Future Work

- Incorporate more years of data when available so the long-term trend can be properly assessed and the deep model provided with more examples of lightning day.
- Instead of an outlook derived from persistence, utilize an actual numerical weather prediction (NWP) stream for both CAPE and humidity.
- Instead of using a district average prediction, focus on a grid-cell prediction as the risk within one district can be highly variable.
- Create a historical view for the dashboard to allow users to contrast the current season's risk for a given district with that from previous years.
- Include more variables identified as relevant by the literature: the K-index and shape of the vertical CAPE profile.

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